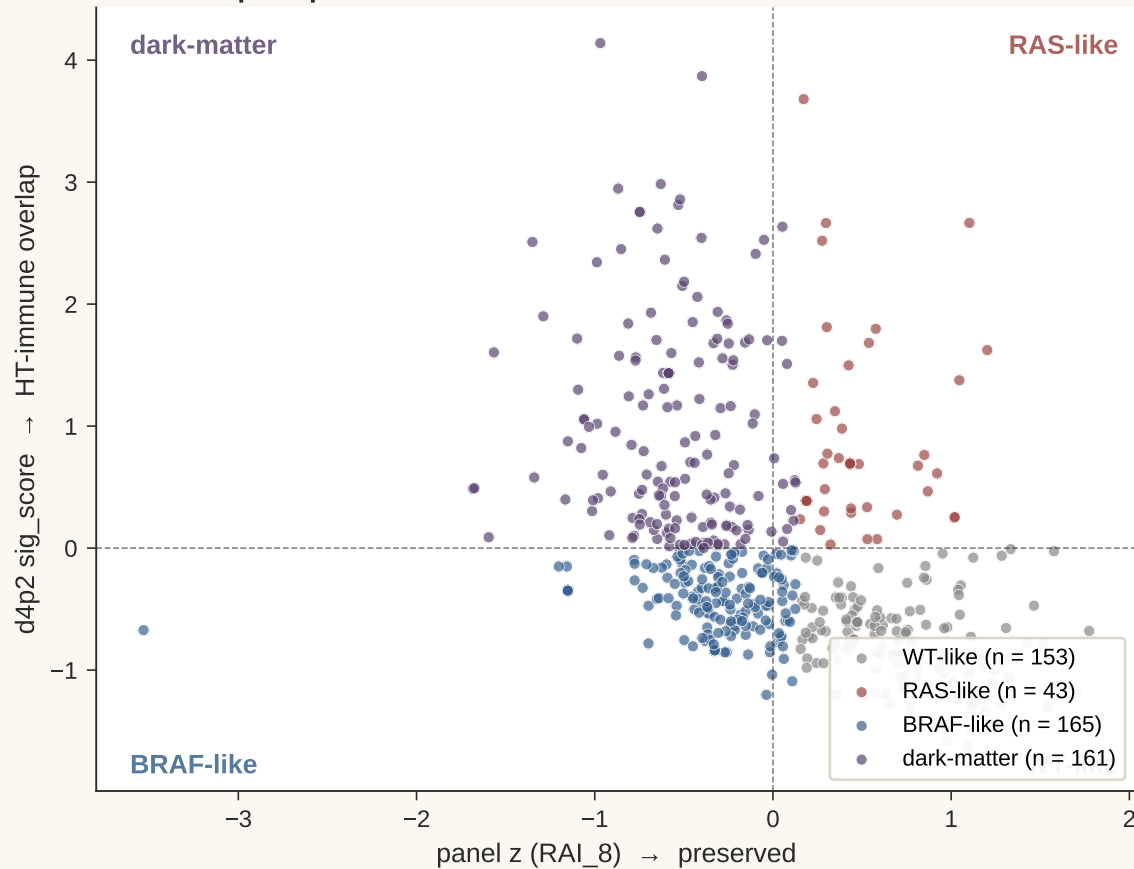


TCGA-THCA two-axis decomposition (n = 522)

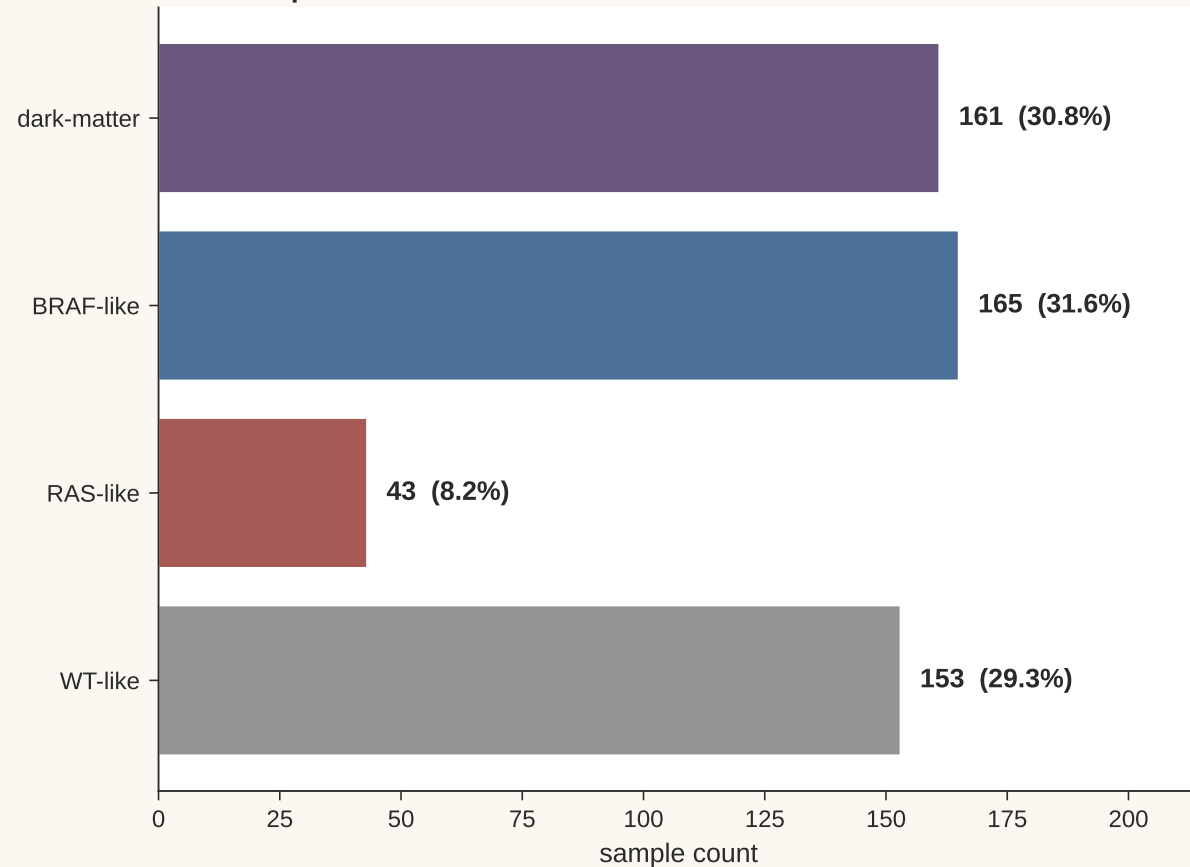
a

Two-axis space partitions 4 zones



b

Zone composition



Two-axis decomposition (panel z × d4p2 HT-overlap) cleanly partitions TCGA-THCA into 4 zones; dark-matter (silenced + HT-overlap) is the highest-risk subgroup.